

Numerical Methods

By

Numerical Methods Group

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Newton's backward interpolation formula

Newton's Backward Interpolation Formula is used to estimate the value of a function when:

- The independent variable values are equally spaced.
- The required value of x is near the end of the table.

It uses backward differences to construct an interpolating polynomial.

Newton's backward interpolation formula

Newton Backward Interpolation Formula:

$$f(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

Where $p = \frac{x - x_n}{h}$

and

x_n is the last value of x in the data table

h = equal spacing

Δy_n = backward difference .

Example :

Calculate the value of $f(84)$ for the data given in the table below:

x	40	50	60	70	80	90
y	204	224	246	270	296	324

Solution :

The value of 84 is near the end of Table. Hence, we use the Newton's backward interpolation formula.

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
40	204					
		20				
50	224		2			
		22		0		
60	246		2		0	
		24		0		0
70	270		2		0	
		26		0		
80	296		2			
		28				
90	324					

Backward difference table (the horizontal table)

We have

$$x = 84, x_n = 90, h = 10$$

$$p = \frac{x - x_n}{h} = \frac{84 - 90}{10} = -0.6$$

Using Newton's backward formula

$$f(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

$$f(84) = 324 + (-0.6)(28) + \frac{(-0.6)(-0.6+1)}{2!} (2) = 306.96$$

Example :

Calculate the value of $f(30)$ for the data given in the table below:

x	21	25	29	33	37
y	18.4708	17.8144	17.1070	16.3432	15.5154

Solution :

The value of 30 is near the end of Table. Hence, we use the Newton's backward interpolation formula.

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
21	18.4708				
		-0.6564			
25	17.8144		-0.0510		
		-0.7074		-0.0054	
29	17.1070		-0.0564		-0.002
		-0.7638		-0.0076	
33	16.3432		-0.0640		
		-0.8278			
37	15.5154				

Backward difference table

We have

$$x = 30, x_n = 37, h = 4$$

$$p = \frac{x - x_n}{h} = \frac{30 - 37}{4} = \frac{-7}{4}$$

Using Newton's backward formula

$$f(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

$$f(84)$$

$$\begin{aligned}
 &= 15.5154 + \left(\frac{-7}{4}\right)(-0.8278) + \frac{\left(\frac{-7}{4}\right)\left(\frac{-7}{4} + 1\right)}{2!}(-0.0640) \\
 &+ \frac{\left(\frac{-7}{4}\right)\left(-\frac{7}{4} + 1\right)\left(-\frac{7}{4} + 2\right)}{3!}(-0.0076) \\
 &+ \frac{\left(-\frac{7}{4}\right)\left(-\frac{7}{4} + 1\right)\left(-\frac{7}{4} + 2\right)\left(-\frac{7}{4} + 3\right)}{4!}(-0.002) = 16.6218
 \end{aligned}$$

Example :

The population (in thousands) of a small town is given in the following table. Estimate the population in the years 1965 and 1995.

x	1961	1971	1981	1991	2001
$f(x)$	46	66	81	93	101

Solution :

Try by yourself

Stirling's interpolation formula

Stirling's interpolation formula is a central difference interpolation formula used to estimate the value of a function when:

- The data points are equally spaced
- The required value is close to the center of the table

It uses a central difference table to construct an interpolating polynomial.

Stirling's interpolation formula

Stirling's interpolation formula:

$$\begin{aligned}
 f(x) &= y_0 + p \frac{\Delta y_0 + \Delta y_{-1}}{2} + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2 - 1)}{3!} \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \\
 &+ \frac{p^2(p^2 - 1)}{4!} \Delta^4 y_{-2} + \dots
 \end{aligned}$$

Where $p = \frac{x - x_0}{h}$

and

x_0 is the central value of x in the data table

h = equal spacing

Δ = forward difference operator.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
$x_{-2} = x_0 - 2h$	y_{-2}				
		Δy_{-2}			
$x_{-1} = x_0 - h$	y_{-1}		$\Delta^2 y_{-2}$		
		Δy_{-1}		$\Delta^3 y_{-2}$	
x_0	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$
		Δy_0		$\Delta^3 y_{-1}$	
$x_1 = x_0 + h$	y_1		$\Delta^2 y_0$		
		Δy_1			
$x_2 = x_0 + 2h$	y_2				

central difference table

Example :

Calculate the value of $f(28)$ for the data given in the table below:

x	20	25	30	35	40
y	48234	47354	46267	44978	43389

Solution :

The value of 28 is near the center of Table. Hence, we use the Stirling interpolation formula.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
$x_{-2} = 20$	48234				
		-880			
$x_{-1} = 25$	47354		-207		
		-1087		5	
$x_0 = 30$	46267		-202		-103
		-1289		-98	
$x_1 = 35$	44978		-300		
		-1589			
$x_2 = 40$	43389				

Central difference table

We have

$$x = 28, x_0 = 30, h = 5$$

$$p = \frac{x - x_n}{h} = \frac{28 - 30}{5} = \frac{-2}{5}$$

Using Stirling formula

$$f(x) = y_0 + p \frac{\Delta y_0 + \Delta y_{-1}}{2} + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2 - 1)}{3!} \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} + \frac{p^2(p^2 - 1)}{4!} \Delta^4 y_{-2} + \dots$$

$$f(28) = 48234$$

$$+ \left(\frac{-2}{5}\right) \frac{(-1087) + (-1289)}{2} + \frac{\left(\frac{-2}{5}\right)^2}{2!} (-202) + \frac{\left(\frac{-2}{5}\right) \left(\left(\frac{-2}{5}\right)^2 - 1\right)}{3!} \frac{(5 - 98)}{2} + \frac{\left(\frac{-2}{5}\right)^2 \left(\left(\frac{-2}{5}\right)^2 - 1\right)}{4!} (-103) = 46724.013$$

Example :

Calculate the value of $f(25)$ for the data given in the table below:

x	10	20	30	40	50
$f(x)$	100	400	900	1600	2500

Solution :

Try by yourself

Lagrange's Formula for Unequal Intervals

Lagrange's Interpolation Formula is used to estimate the value of a function when:

- The independent variable values are not necessarily equally spaced.

Suppose we are given $n + 1$ data points:

$$(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

where

$$y_i = f(x_i)$$

We want to find the value of $f(x)$ for some value of x .

Lagrange assumed that the function can be approximated by a polynomial of degree n passing through all these points.

Lagrange's Interpolation Formula

$$f(x) = \sum_{i=0}^n L_i(x) y_i, \quad L_i(x) = \prod_{\substack{j=0 \\ i \neq j}}^n \frac{x - x_j}{x_i - x_j}$$

The formula can be written in the form:

$$\begin{aligned} f(x) &= \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 \\ &+ \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1 + \dots \\ &+ \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} y_n \end{aligned}$$

Example :

Apply Lagrange's interpolation formula to find a polynomial which passes through the points $(0, -20)$, $(1, -12)$, $(3, -20)$ and $(4, -24)$.

Solution :

We have $x_0 = 0, x_1 = 1, x_2 = 3, x_3 = 4, y_0 = f(x_0) = -20, y_1 = f(x_1) = -12, y_2 = f(x_2) = -20$ and $y_3 = f(x_3) = -24$.

The Lagrange's interpolation formula is:

$$\begin{aligned} f(x) &= \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} y_0 + \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} y_1 \\ &+ \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} y_2 + \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)} y_3 \end{aligned}$$

$$\begin{aligned} & f(x) \\ &= \frac{(x-1)(x-3)(x-4)}{(0-1)(0-3)(0-4)}(-20) + \frac{(x-0)(x-3)(x-4)}{(1-0)(1-3)(1-4)}(-12) \\ &+ \frac{(x-0)(x-1)(x-4)}{(3-0)(3-1)(3-4)}(-20) + \frac{(x-0)(x-1)(x-3)}{(4-0)(4-1)(4-3)}(-24) \\ &= x^3 - 8x^2 + 15x + 20 \end{aligned}$$

Example :

Apply Lagrange's interpolation formula to find a polynomial which passes through the points $(0, -12)$, $(1, 0)$, $(3, 6)$ and $(4, 12)$.

Solution :

Try by yourself

Example :

Using Lagrange's interpolation formula, find the value of y corresponding to $x = 10$ from the following data.

x	5	6	9	11
y	380	-2	196	508

Solution :

The Lagrange's interpolation formula is:

$$\begin{aligned}
 f(x) &= \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} y_0 + \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} y_1 \\
 &+ \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} y_2 + \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)} y_3
 \end{aligned}$$

$$x = 10, x_0 = 5, x_1 = 6, x_2 = 9, x_3 = 11$$

$$y_0 = 380, y_1 = -2, y_2 = 196, y_3 = 508$$

$$f(10)$$

$$\begin{aligned} &= \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)}(380) + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)}(-2) \\ &+ \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)}(196) + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)}(508) = 330 \end{aligned}$$

Lagrange's Formula for Inverse Interpolation

In Lagrange interpolation formula y is expressed as a function of x as:

$$y = \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 + \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1 + \dots + \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} y_n$$

By interchanging x and y , we can express x as a function of y as follows:

$$x = \frac{(y - y_1)(y - y_2) \dots (y - y_n)}{(y_0 - y_1)(y_0 - y_2) \dots (y_0 - y_n)} x_0 + \frac{(y - y_0)(y - y_2) \dots (y - y_n)}{(y_1 - y_0)(y_1 - y_2) \dots (y_1 - y_n)} x_1 + \dots + \frac{(y - y_0)(y - y_1) \dots (y - y_{n-1})}{(y_n - y_0)(y_n - y_1) \dots (y_n - y_{n-1})} x_n$$

The above equation can be used for inverse interpolation.

Example :

Find the value of x when $y = 250$ by applying Lagrange's inverse interpolation

x	1	2	5	7
y	1	12	117	317

Solution :

The Lagrange's interpolation formula is:

$$\begin{aligned}
 x &= \\
 &= \frac{(y - y_1)(y - y_2)(y - y_3)}{(y_0 - y_1)(y_0 - y_2)(y_0 - y_3)} x_0 + \frac{(y - y_0)(y - y_2)(y - y_3)}{(y_1 - y_0)(y_1 - y_2)(y_1 - y_3)} x_1 \\
 &+ \frac{(y - y_0)(y - y_1)(y - y_3)}{(y_2 - y_0)(y_2 - y_1)(y_2 - y_3)} x_2 + \frac{(y - y_0)(y - y_1)(y - y_2)}{(y_3 - y_0)(y_3 - y_1)(y_3 - y_2)} x_3
 \end{aligned}$$

$$x_0 = 1, x_1 = 2, x_2 = 5, x_3 = 7$$

$$y = 250, y_0 = 1, y_1 = 12, y_2 = 117, y_3 = 317$$

$$f(10)$$

$$= \frac{(250 - 12)(250 - 117)(250 - 317)}{(1 - 12)(1 - 117)(1 - 317)} (1)$$

$$+ \frac{(250 - 1)(250 - 117)(250 - 317)}{(12 - 1)(12 - 117)(12 - 317)} (2) + \frac{(250 - 1)(250 - 12)(250 - 317)}{(117 - 1)(117 - 12)(117 - 317)} (5)$$

$$+ \frac{(250 - 1)(250 - 12)(250 - 117)}{(317 - 1)(317 - 12)(317 - 117)} (7) = 6.7$$

Example :

Find the value of x when $y = 420$ by applying Lagrange's inverse interpolation

x	1	2	5	6
y	71	82	295	466

Solution :

Try by yourself

**THANK YOU
FOR YOUR
ATTENTION**

